



Figure 3: The ns-3 network simulator

- Multi-level connectivity with LTE

The mmWave module is integrated with ns-3, which has to be installed along with the ns-3 network simulator so that 5G network simulation features can be integrated with the existing features. To work with the mmWave module in the ns-3 network simulator with LTE support, the following instructions are executed:

```
$ sudo apt install git build-essential (Activation of the
Repository and the Packages)

$ git clone https://github.com/nsnam/ns-3-dev.git ns-3-dev
(Cloning and Fetching the Library of ns-3)
$ cd ns-3-dev/src (Moving to the SRC Directory in ns-3)
$ git clone https://github.com/nyuwireless-unipd/mmwave
mmwave (Activation, Cloning and Fetching the mmWave with ns3)
$ cp mmwave/ns-3-patches/patch-for-lte.diff lte
(Initialization of LTE)
$ cd lte
$ patch -p1 -N -E -i patch-for-lte.diff
$ rm patch-for-lte.diff
```

To configure and build ns-3 from the *ns-3-dev* folder, the following command is used:

```
$ ./waf configure --enable-tests --enable-examples
$ ./waf build
```

To integrate the Python programming, the following command is used:

```
$ ./waf configure --enable-tests --enable-examples --disable-
python
$ ./waf build
```

The mmWave module is available with the ns-3 network simulator so that the simulation of the 3rd Generation Partnership Project (3GPP) cellular network scenarios can be simulated to test multiple factors and parameters. The ns-3 network simulator is available at the official link <https://www.nsnam.org/> that provides the implementation patterns of advanced and next generation networks with the integration of numerous protocols and algorithms.

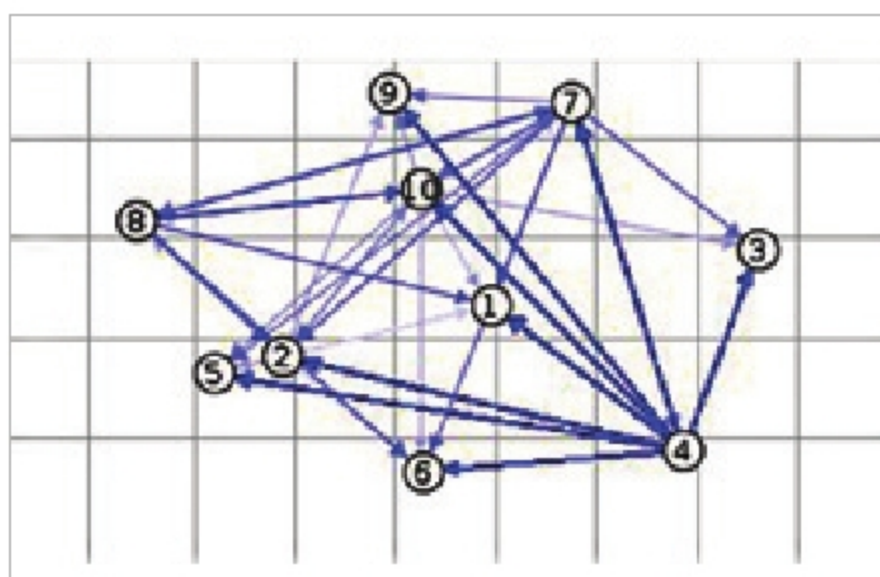


Figure 4: Simulating a network environment

The installation of ns-3 can be done on Linux operating systems. On Windows, there is need to install ns-3 using virtual machines. The ns-3 network simulator is a widely used discrete event simulator for network environments. Earlier, the use of the ns-2 network simulator was quite common, but there were many limitations in implementing recent protocols and algorithms. The ns-3 mmWave module simulates Long Term Evolution (LTE) based networks, so that the performance and behaviour of networks in 5G and upcoming generations can be predicted.

Using the ns-3 network simulator, a mobile network can be formed with mobile nodes in 4G, 5G or similar network environments. With the integration of the graphical user interface (GUI) in ns-3, the formation and behaviour of the network can be viewed. A number of attacks can be simulated using mmWave with ns-3. The following are some of the network assaults that can be simulated for different scenarios with multiple mobile nodes on cellular networks.

Mobile node imitation and the Sybil attack: In this type of network assault, similar types of nodes are created to disguise the servers. Using mmWave ns-3, the simulation of Sybil or node imitation attacks can be done so that any similar future attempt on a 5G network can be avoided.

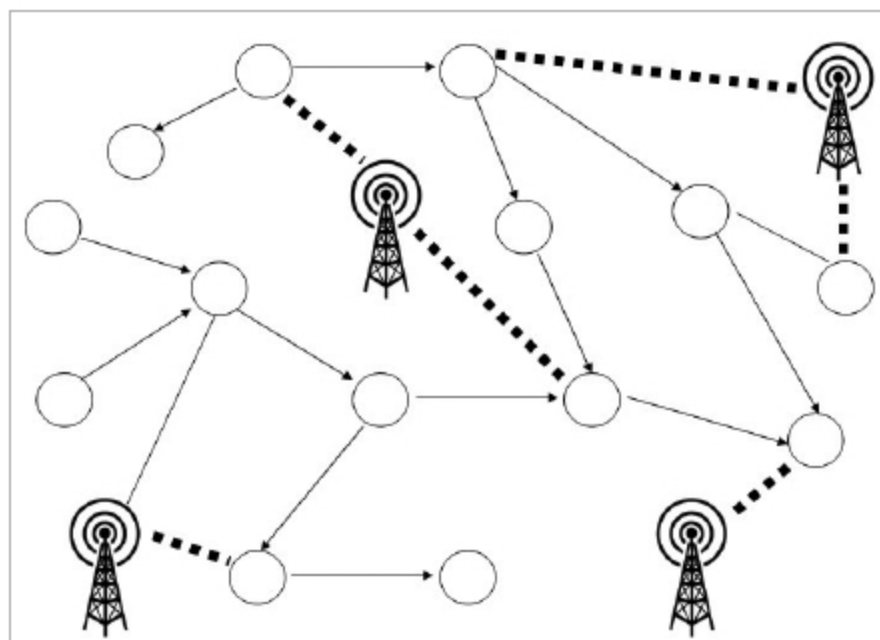


Figure 5: Integrating cellular network base stations and the related nodes